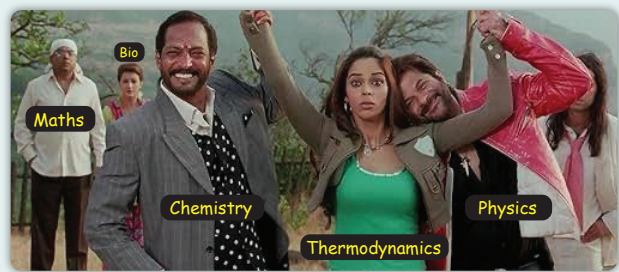


One of the most scoring chapter KTG and thermodynamics.... in all format of exam Advance/Mains/NEET..... Jitna main yahaan cover kar raha hoon, utna agar aapne digest kar liya to is chapter ke PYQ question ke solve hone ki possibility 90% tak hai. Bas yahaan aapko kuch formula yaad rakhne hain aur unko apply karna seekhna hai, although aap ise chemistry mein bhi padh chuke hain so uski help yahaan milegi....



KTG AND THERMODYNAMICS

Assumption of Kinetic Theory of Gas

- ★ All gas particles are moving in random direction.
- ★ There is no attraction between any two particles.
- ★ There is no repulsion between any two particles.
- ★ There is no interaction between any two particles.
- ★ Potential energy of system = 0
- ★ Total energy will be in the form of Kinetic Energy.
- ★ All the collision are elastic in nature. (particle-particle) and (particle-wall)
- ★ Effect of gravity is neglected.
- ★ Volume of particle is negligible with respect volume of container.
- ★ Real gas will behave like ideal gas at High temperature & low pressure.

IMPORTANT FORMULAS

Average Velocity

$$\text{Avg. Velocity} = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \dots + \vec{v}_n}{n} = 0$$

where $\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_n$ are velocities of first second third... n^{th} particle respectively.

Average Speed

It is the average of speed of all particle it is given by

$$\text{Avg Speed} = \sqrt{\frac{8}{\pi} \frac{RT}{M}}$$

Where R is universal gas constant T is temperature in kelvin M is molar mass

Most Probable Speed

It is speed of maximum fraction of gas molecules it is given by.

$$\text{Most prob. speed}(V_{m.p}) = \sqrt{\frac{2RT}{M}}$$

Root Mean Square Speed (RMS Speed)

RMS speed is square root of the average of square of the velocity of individual gas molecule. It is given by

$$(\text{Root Mean sq. vel.}) V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$V_{rms} > V_{avg} > V_{m.p}$$

Q. If a particle have a set of velocities 5,3,0,3,3,2,5,1,4,2,m/s, then we take mean of all speed

Sol. Average speed mean of all speeds

$$\text{Avg. Speed} = \frac{5+3+0+3+3+2+5+1+4+2}{10} = 2.8$$

Root mean square speed is calculated as

$$V_{\text{rms}} = \sqrt{\frac{5^2 + 3^2 + 0^2 + 3^2 + 3^2 + 2^2 + 5^2 + 1^2 + 4^2 + 2^2}{10}}$$

$$= \sqrt{10 \cdot 2}$$

Most probable speed:

$V_{\text{mp}} = 3$ (Most particle have velocity 3 m/s)

Q. Find rms velocity (v_{rms}) of oxygen molecule at 27°C

Sol. RMS velocity is given by

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 25 \times 300}{3 \times 32 \times 10^{-3}}}$$

Ideal Gas

Gas which follows ideal gas equation is called ideal gas. Real gases obey ideal gas equation at high temperature and low pressure

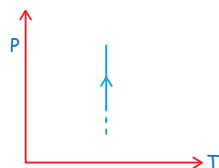
$$PV = nRT \text{ (ideal gas equation)}$$

$$PM = \rho RT$$

Abe chemistry mein padha tha, itna toh yaad hai na

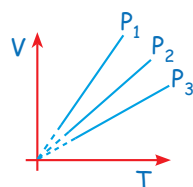
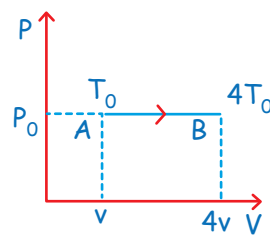
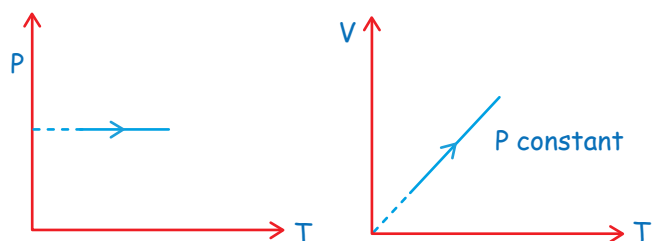
BOYLE'S LAW (ISOTHERMAL PROCESS)

- ★ Temp constant
- ★ $PV = \text{constant}$



CHARLE'S LAW (ISOBARIC PROCESS)

- ★ Pressure constant
- ★ $V \propto T$



$$P_1 < P_2 < P_3$$

$$\text{Slope} = \frac{nR}{P}$$

Slope $\uparrow \Rightarrow P \downarrow$

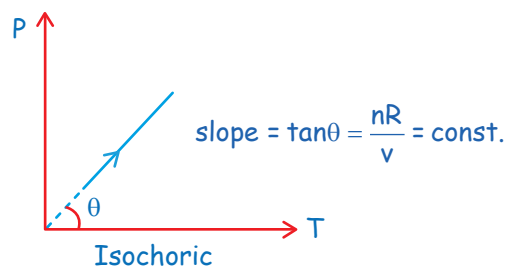
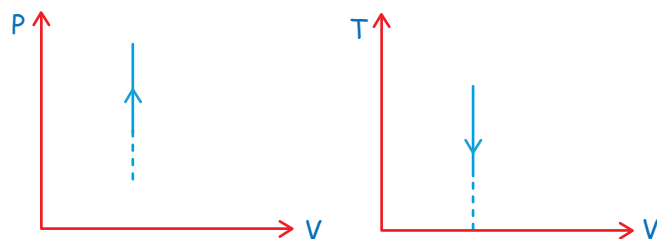
SKC

Thermodynamics ka sabse bada hathiyar (weapon) hai... PV ka product jitna times ho jaayega, temp utne hi times ho jaayega (by keeping n constant)

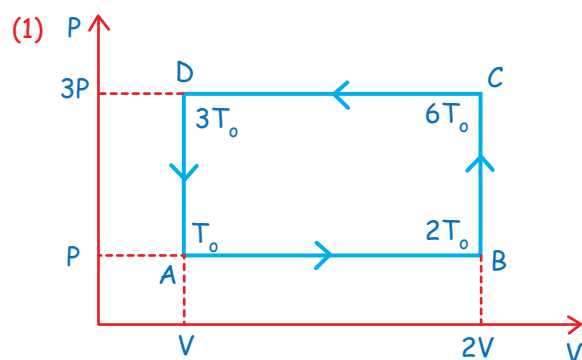


GAY LUSSAC'S LAW (ISOCORIC PROCESS)

- ★ Volume constant
- ★ $P \propto T$

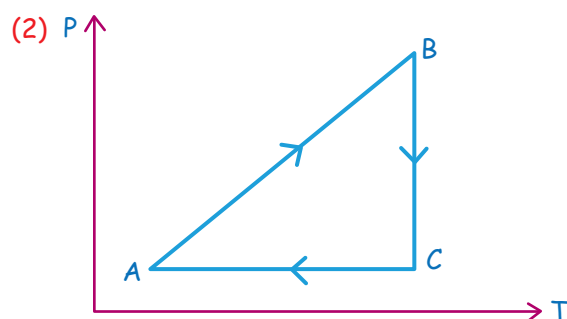


Q. An ideal gas undergoes following thermodynamic process as shown in figure. Discuss changes in pressure, volume and temperature.



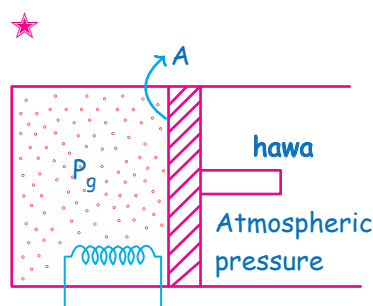
Sol.

	P	V	T	Process
A → B	Const.	↑	↑	Isobaric
B → C	↑	Const.	↑	Isochoric
C → D	Const.	↓	↓	Isobaric
D → A	↓	Const.	↓	Isochoric

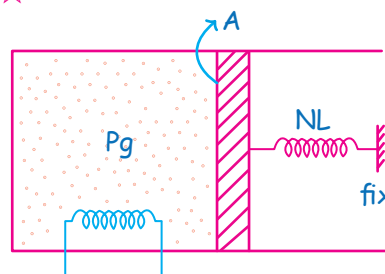


Sol. In process A to B volume is constant so it is isochoric process & pressure increase on increasing temperature. In B to C temperature is constant so it is isothermal process, pressure decrease and volume increase. In process C to A, pressure is constant so it is isobaric process, so temperature and volume both decrease.

	P	V	T	Process
A → B	↑	Const.	↑	Isochoric
B → C	↓	↑	Const.	Isothermal
C → A	const.	↓	↓	Isobaric

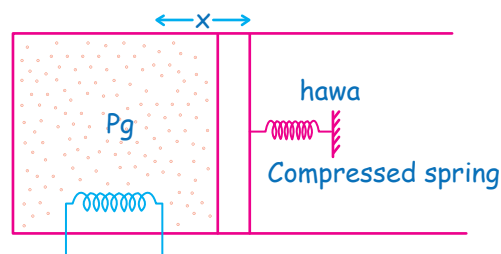


As heater switch on, gas expand and volume of gas increases, piston start moving slowly to right and process will be isobaric.

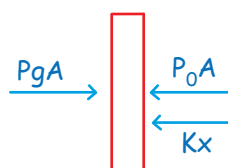


Volⁿ increasing
n → same

As heater switch on, due to heating Piston moves by x and spring is now compressed by x.



Forces acting on the piston are as shown in the figure



Ye isobaric nahi hai.
Galti mat kar
dena warna pitaai
hogi

Balancing forces, we get

$$P_g A = P_0 A + Kx$$

$$P_g = \frac{P_0 A + Kx}{A}$$

$$P_g = P_0 + \frac{Kx}{A}$$

DEGREE OF FREEDOM (DOF)

No. of independent way in which a molecule can exhibit motion or have energy is called Degree of Freedom (Dof)

Monoatomic Gas

Eg: (all inert gases, He, Ar, etc.)

$$f = 3 \text{ (translational)}$$

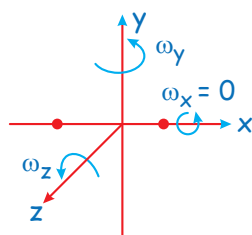
$$(V_x, V_y, V_z)$$

Diatomic Gas

Eg: (gases like H₂, N₂, O₂ etc.)

$$f = 5 \text{ (3 translational + 2 rotational)}$$

$$(V_x, V_y, V_z; \omega_y, \omega_z)$$



Note: For diatomic gas:

- (a) At low temperature
 $f = 3$ (translational)
- (b) At room temperature
 $f = 5$ (3 translational + 2 rotational)
- (c) At high temperature
 $f = 7$ [3 translational + 2 rotational + 2 vibrational]

Triatomic (Non-linear) Gas

$f = 6$ [3 translational + 3 rotational]

For example: H_2O

Triatomic (linear) Gas

$f = 5$ [3 translational + 2 rotational]

For example: CO_2

MAXWELL'S EQUIPARTITION LAW

Energy corresponding to each degree of freedom of an ideal gas molecule = $\frac{1}{2} kT$
where k is Boltzmann constant T is temperature of gas

so energy correspond to one molecule of gas of def. 'f' = $fkT/2$.

$$k = R/N_A$$

Total energy of 1 molecule (U) = $\frac{1}{2} kfT$

Total energy of 1 mole of gas = $N_A/2 fkT = fRT/2$

$$\text{Total energy of } n \text{ mole} = \frac{nfRT}{2}$$

$n \rightarrow$ no. of moles

$f \rightarrow$ dof

$T \rightarrow$ Temp (K)

$U \rightarrow$ Internal energy

$U \rightarrow$ depends on 'T'

This total energy (kinetic energy) is also called

internal energy $U = \frac{nfRT}{2}$

$$\rightarrow U = \frac{nfRT}{2} \quad U \propto T$$

If Temperature increases internal energy also increase

Change in internal energy of gas can be given by

$$\Delta U = \frac{nfR\Delta T}{2}$$

If temperature increase change in internal energy is positive and if temperature decrease change in internal energy is negative

$$T \uparrow \Rightarrow U \uparrow \Rightarrow \Delta U > 0$$

$$T \downarrow \Rightarrow U \downarrow \Rightarrow \Delta U < 0$$

Q. A container x has volume double that of container y and both are connected by a thin tube Both contain same ideal gas. The Temp of x is 200 K and that of y is 400 K. If mass of gas in x is m then in y it will be:

Sol.	X	Y	ideal gas eq ⁿ for y
	2V	V	$y = PV = \frac{m'}{M} R \cdot 400$
	200K	400K	Ideal gas eq ⁿ for x
	P	P	$x = P2V = \frac{m}{M} R \cdot 200$
	m	m'	Solving two eq ⁿ
			$2 = \frac{m}{m'} \times \frac{1}{2}$

we get

$$m' = m/4$$

Q. At a Temp T 'K'. the pressure of 4.0 g argon in a bulb is P. The bulb is put in a bath having temp higher by 50 K than the first one 0.8 g of argon gas had to be removed to maintain original pressure. The temp. T is equal to-

Sol. $PV = nRT$

$$\text{Initially } PV = 4/40 RT \quad \dots(1)$$

After 0.8g argon is removed

$$PV = 3.2/40 R(T + 50) \quad \dots(2)$$

$$\frac{4}{40} RT = \frac{3.2}{40} R(T + 50)$$

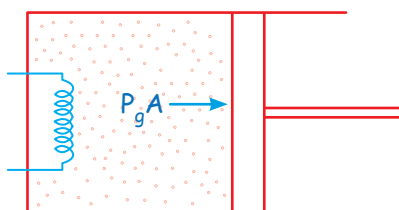
$$\frac{40}{32} T = T + 50$$

$$\frac{10T}{8} = T + 50$$

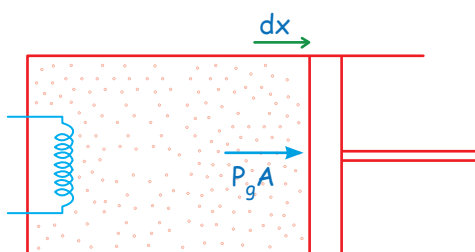
$$10T = 8T + 400$$

$$T = 200$$

WORK DONE BY THE GAS (W_g)



suppose piston moves by dx
work done by gas = $\int F dx = \int P_g A dx = \int P_g dv$

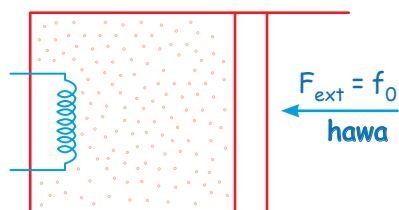


$$(WD)_{\text{gas}} = \int P dv$$

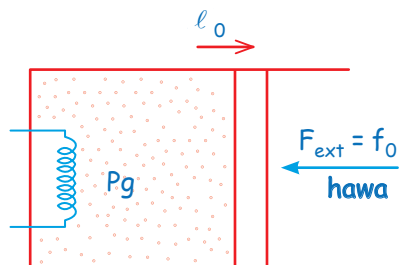
Pressure of the gas

- ★ If $V \uparrow \Rightarrow$ gas expand $\Rightarrow w_g > 0$
If volume increases work done by gas > 0
- ★ If $V \downarrow \Rightarrow$ gas compress $\Rightarrow w_g < 0$
If volume decreases work done by gas < 0
- ★ If $V \rightarrow \text{const} \Rightarrow w_g = 0$
If volume is constant work done by gas $= 0$

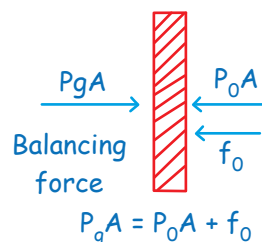
Q. Find work done by gas against external force f_0 and atmospheric pressure P_0 which is heated as shown in fig. Piston moves ℓ_0 distance. (Given cross sectional area of piston is A)



Sol. Let P_g be the pressure inside gas



work done by gas (W_g) = $\int P_g dv = \int P_g A dx$

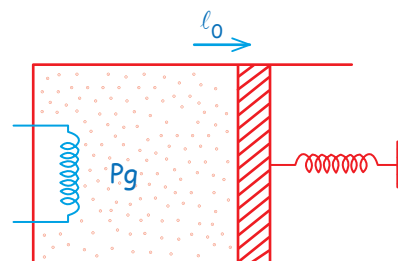
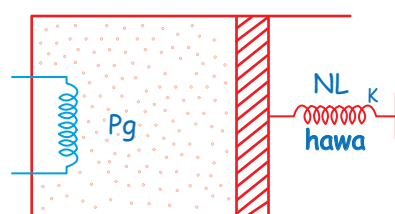


$$P_g A = P_0 A + f_0$$

$$W_g = \int_0^{\ell_0} (P_0 A + f_0) dx = (P_0 A + f_0) \int_0^{\ell_0} dx$$

$$= (P_0 A + f_0) \ell_0$$

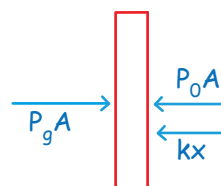
Q. Spring is attached to piston as shown in figure on one end and to a rigid support on other end. spring was in natural length initially. Piston moves slowly by ℓ_0 . Find work done by gas



Haule-Haule

Sol. Work done by $_{\text{gas}} = \int P_g dv$

$$W_g = \int P_g A dx$$



$$P_g A = P_0 A + Kx \text{ (Force balance)}$$

$$W_g = \int_0^{\ell_0} (P_0 A + Kx) dx = P_0 A \ell_0 + \frac{K \ell_0^2}{2}$$

Method 2

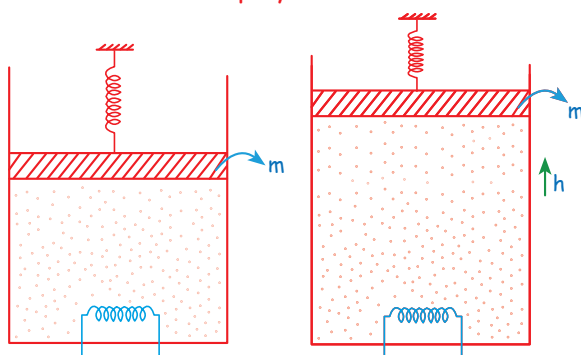
Work energy theorem-

$$(W)_g + (W)_{\text{hawa}} + (W)_{\text{sp}} = \Delta K.E$$

$$(W)_g - P_0 A \ell_0 - 1/2 K (\ell_0^2 - 0^2) = 0$$

$$(WD)_g = P_0 A \ell_0 + \frac{K \ell_0^2}{2}$$

Q. Vertical cylinder is shown in the following diagram. One end of the spring is attached to the piston and the other to the rigid support. Initially, the spring is in its natural length. Find the work done by the gas. The piston of mass m moves up by h .



Sol. Work energy theorem-

$$W_{\text{gas}} + W_{\text{hawa}} + W_{\text{sp}} + W_{\text{gravity}} = \Delta K.E$$

$$W_{\text{gas}} - P_0 A h - \frac{1}{2} K (h^2 - 0^2) - mgh = 0$$

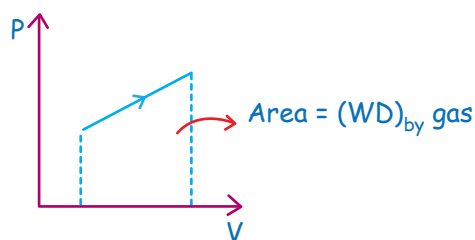
$$W_g = P_0 A h + \frac{1}{2} K (h^2 - 0^2) + mgh$$

$$= P_0 A h + \frac{1}{2} K h^2 + mgh$$

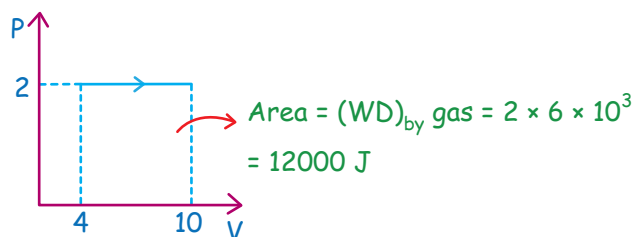
★ **Work done by gas = Area under P-V curve**

$$(WD)_{\text{gas}} = \int P dV$$

$$= \text{Area}$$



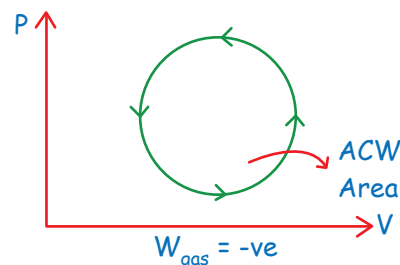
Q. Find W_{gas} in the following P-V diagram



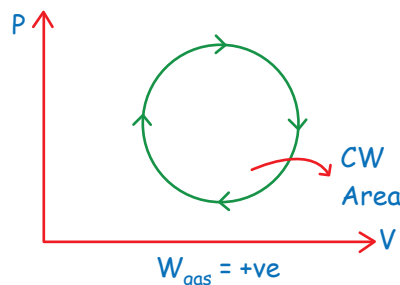
CYCLIC PROCESS

In a cyclic process, a system undergoes a series of changes and returns to its initial state, meaning the internal energy change is zero, and the work done equals the heat absorbed.

In an anti-clockwise thermodynamic process $W_{\text{gas}} = -ve$



In a clockwise thermodynamic process, the work done by the gas is +ve.



Cyclic Process mein $\Delta U_{\text{net}} = 0$, change in internal energy = 0



Kaam ka Dabba

Dekh bhai ab tak jo humne padha hai uska nichod (summary/essence) ye hai. Tu ise catch kar

Mono atomic dof = 3

Diatomic $f = 5$ (at high temp $f = 7$)

Triatomic linear $f = 5$

Triatomic non-linear $f = 6$

$$(WD)_{\text{gas}} = \int P dV$$

$$\text{If } V \uparrow \Rightarrow (WD)_{\text{gas}} > 0$$

$$\text{If } V \downarrow \Rightarrow (WD)_{\text{gas}} < 0$$

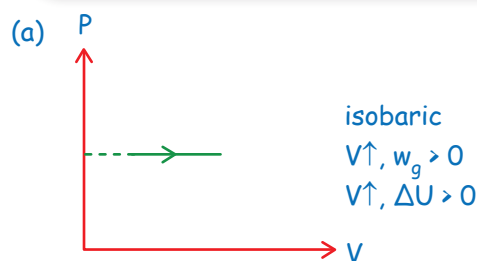
$$\Delta U = \frac{nfR\Delta T}{2}$$

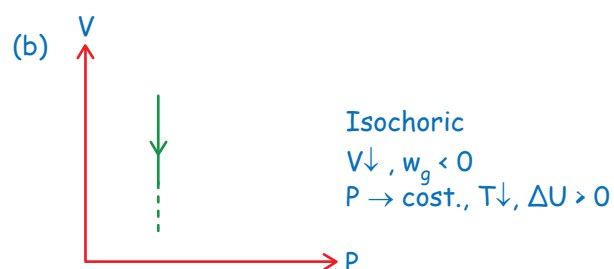
$$\text{If } T \uparrow \Rightarrow \Delta U > 0$$

$$\text{If } T \downarrow \Rightarrow \Delta U < 0$$

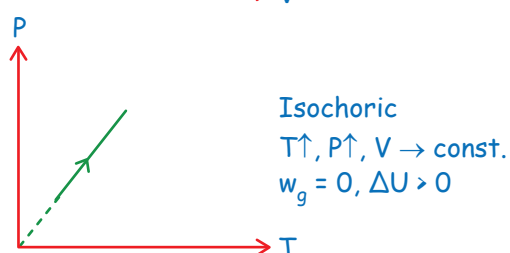
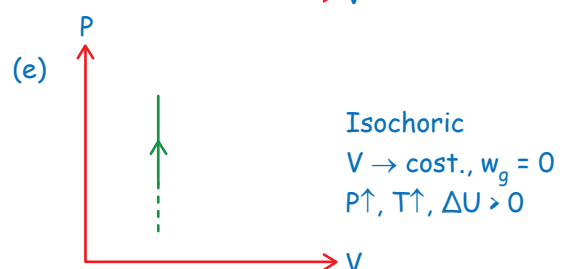
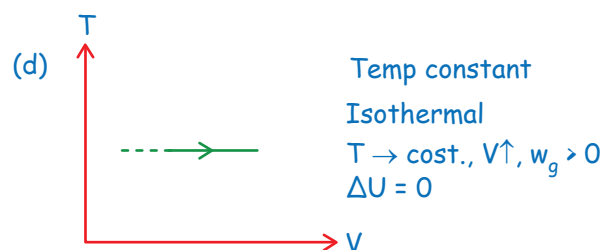
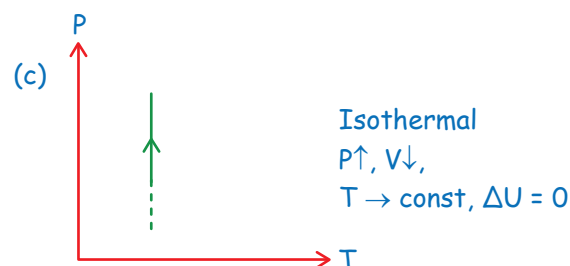
★ P-V graph ka area $(WD)_{\text{by gas}}$ dega

★ Cyclic Process mein $\Delta U_{\text{net}} = 0$, change in internal energy = 0

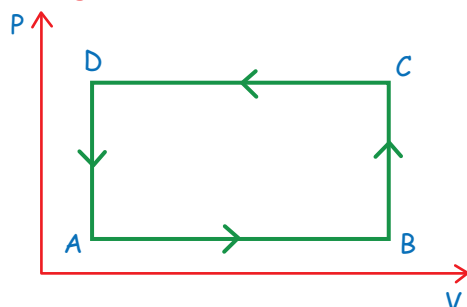




If Vol^m decrease, pressure also decrease.

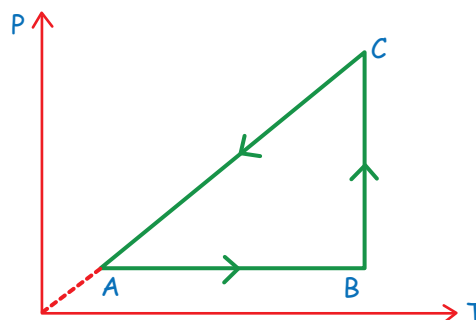


Q. In following thermodynamic process P-V diagram, comment about the sign of work done by gas change in internal energy, process and heat given.



	W_g	ΔU	Process	ΔQ
$A \rightarrow B$	+	+	Isobaric	+
$B \rightarrow C$	0	+	Isochoric	+
$C \rightarrow D$	-	-	Isobaric	-
$D \rightarrow A$	0	-	Isochoric	-

Q. In following thermodynamic process, find sign of work done by gas and internal energy change in each process A to B, B to C, C to A

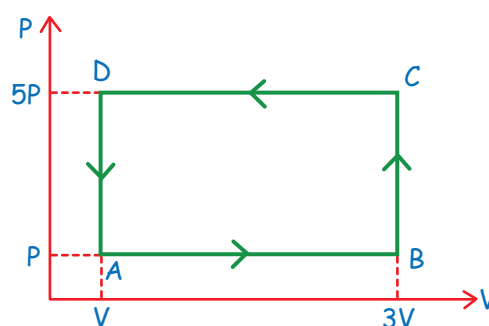


Sol. Process AB is isobaric process BC isothermal and process CA is isochoric.

In process AB temperature and volume increase so work done by gas and internal energy change is positive. In process BC temperature is constant pressure increase and volume decrease so, work done by gas is negative and internal energy change is 0. In process CA volume is constant pressure decrease and temperature also decrease. so work done by gas is zero and internal energy change is negative.

	W_g	ΔU
$A \rightarrow B$	+	+
$B \rightarrow C$	-	0
$C \rightarrow A$	0	-

Q. In the following thermodynamics process to moles of mono atomic gas under goes process $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ given temperature at $A = 100\text{K}$ and degree of freedom of gas = 3



Find net work done & net change in internal energy

Sol. Given $T_A = 100\text{K}$, From ideal gas eqⁿ $PV = nRT$
Temperatures at B, C and D are respectively $T_B = 300\text{K}$, $T_C = 1500\text{K}$, $T_D = 500\text{K}$

Work done by gas from A to B = area under AB
 $P(3V - V)$

$$= +2PV$$

work done by gas from B to C = 0 (isochoric process) = work done by gas from D to A

$$\text{work done by gas C to D} = -5P \times 2V = -10PV$$

Net work done by gas =

$$W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$2 + 0 - 10 + 0$$

$$= -8PV$$

$$T_A = 100\text{K}, n = 2, f = 3$$

Change in internal energy can be found by putting values of T, n, f in the formula

$$= \frac{nfR\Delta T}{2}$$

$$\Delta U_{A \rightarrow B} = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(300 - 100)}{2} = +600R$$

$$\Delta U_{B \rightarrow C} = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(1200)}{2} = +3600R$$

$$\Delta U_{C \rightarrow D} = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(-1000)}{2} = -3000R$$

$$\Delta U_{D \rightarrow A} = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(-400)}{2} = -1200R$$

$$\Delta U_{\text{Net}} = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CD} + \Delta U_{DA}$$

	(WD) _g	ΔU
A $\rightarrow$ B	+2PV	+600 R
B $\rightarrow$ C	0	+3600 R
C $\rightarrow$ D	-10PV	- 3000 R
D $\rightarrow$ A	0	- 1200 R
Net	-8 PV	0

FIRST LAW OF THERMODYNAMICS

★ Based on conservation of energy

$$\Delta Q = \Delta U + \Delta W$$

$\Delta U \rightarrow$ Change in Internal Energy

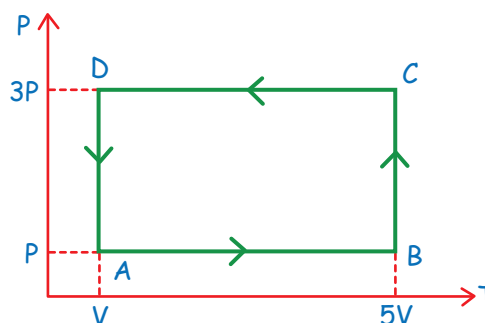
$\Delta W \rightarrow$ (WD) by gas.

$\Delta Q \rightarrow$ Heat given/taken

Sign Convention

- ★ $\Delta Q \Rightarrow$ Heat given to system $\Delta Q > 0$
- ★ Heat taken away from system $\Delta Q < 0$
- ★ $\Delta U \Rightarrow$ If temperature increase change in internal energy $\Delta U > 0$
- ★ If temperature decrease change in internal energy, $\Delta U < 0$
- ★ $\Delta W \Rightarrow$ If volume increase work done by gas is positive and if volume decrease work done by gas is negative

Q. In following thermodynamic process find the sign of work done ΔW , change in internal energy ΔU , heat supplied ΔQ



Sol.

	ΔW	ΔU	$\Delta Q = \Delta U + \Delta W$
A $\rightarrow$ B	+	+	+
B $\rightarrow$ C	0	+	+
C $\rightarrow$ D	-	-	-
D $\rightarrow$ A	0	-	-

For thermodynamic process $PV^2 = \text{constant}$ Process

If volume become double,

P becomes 1/4 times

T becomes 1/2 times

Temp dekhne ke liye ye dekho ki PV ka product kitne times hua hai

Q. For $P = KV^2$ - thermodynamic Process 2 moles monoatomic gas at temperature 100 K, initial volume $V_i = 2\text{m}^3$ If volume become double, Find work done by gas change in internal energy and heat supplied.

Sol. $P = KV^2$

If $V \rightarrow \text{double}$

P becomes 4 times

T becomes 8 times

= 800 K

Work done by gas

$$W_g = \int PdV = \int_2^4 KV^2 dV = K \left(\frac{4^3}{3} - \frac{2^3}{3} \right) = \frac{56K}{3}$$

Change in internal energy

$$\Delta U = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times 25 \times (700)}{3 \times 2} = 700 \times 25 = 17500$$

$$\Delta Q = \Delta U + \Delta W$$

Q. For thermodynamic process $P = KV^3$, 2 mole of mono atomic gas has initial temperature $T_i = 100$ K if volume changes from $(V_0) \rightarrow$ to $(4V_0)$ Find work done by gas and change in internal energy.

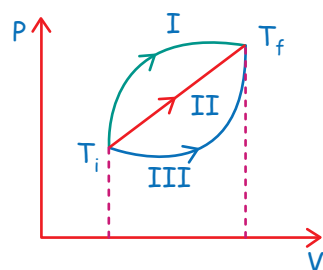
Sol. Change in internal energy

$$\Delta U = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R \times (25600 - 100)}{2}$$

Work done by gas

$$(WD)_{\text{gas}} = \int_{V_0}^{4V_0} KV^3 dV = \frac{K255V_0^4}{4}$$

Q. An ideal gas reaches from initial temp T_i to final temp T_f from paths I, II & III. Compare work done W_g change internal energy ΔU and heat ΔQ supplied, along different paths.



Sol. Internal energy change all paths are same as it is path independent

$$\Delta U \Rightarrow \Delta U, = \Delta U_2 = \Delta U_3$$

Work done is = area under curve so

$$W_g = I > II > III$$

From first law thermodynamic

$$\Delta Q = \Delta W + \Delta U$$

So heat supplied in path $I > II > III$

WORK DONE BY GAS IN DIFFERENT PROCESSES

Isochoric Process

V remains constant so work done by gas = 0

From first law of thermodynamic

$$\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta U$$

Isobaric Process

Pressure remains constant

work done by gas (volume changes from V_i to V_f)

$$w_g = \int PdV = P \int_{V_i}^{V_f} dV = P(V_f - V_i)$$

If temperature changes by ΔT

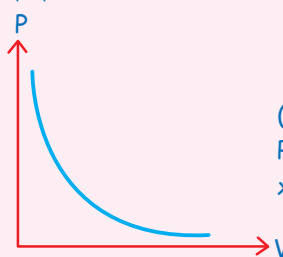
$$w_g = P\Delta V = nR\Delta T$$

Isothermal Process

★ Temperature constant $\Rightarrow \Delta U = 0$

$PV \rightarrow \text{const.}$

$$P_i V_i = P_f V_f$$



(Isothermal)
 $PV = nRT = \text{const}$
 $xy = \text{const.}$

★ $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta W$

★ Work done by gas when volume changes from V_i to V_f

$$w_g = \int PdV = \int \frac{nRT}{V} dV = nRT \int_{V_i}^{V_f} \frac{dV}{V}$$

$$w_g = nRT \ln \frac{V_f}{V_i} = nRT \ln \left(\frac{P_i}{P_f} \right)$$

Adiabatic Process

$PV^\gamma = \text{Constant}$, (where γ is an adiabatic constant)
No heat or energy is exchanged between the system and its surroundings.

$$\Delta Q = 0$$

$$\Rightarrow \Delta W = -\Delta U$$

Work done by gas is given by

$$(W_D)_{\text{gas}} = \frac{nR\Delta T}{1-\gamma} = \frac{P_2V_2 - P_1V_1}{1-\gamma}$$

↙ constant

Proof

$$PV^\gamma \rightarrow \text{const}$$

If gas expand adiabatically from $V_1 \rightarrow V_2$

$$P_1V_1^\gamma = P_2V_2^\gamma = K \text{ (Let)}$$

work done by gas

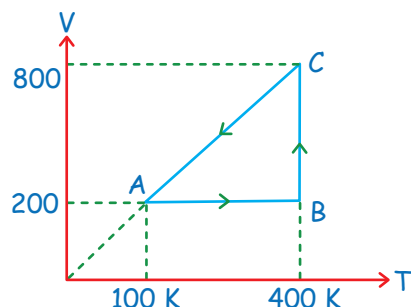
$$(W_D)_g = \int Pdv = \int_{V_1}^{V_2} \frac{K}{v^\gamma} dv = K \int_{V_1}^{V_2} v^{-\gamma} dv$$

$$W_g = K \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2} = \frac{K}{1-\gamma} (V_2^{1-\gamma} - V_1^{1-\gamma})$$

$$W_g = \frac{P_2V_2 - P_1V_1}{1-\gamma} = \frac{nR\Delta T}{1-\gamma} \text{ (Already solve)}$$

$$W_g = \frac{nR\Delta T}{1-\gamma}$$

Q. 2 mole monoatomic gas undergoes thermodynamic process as shown in figure. Find work done by gas change in internal energy heat supplied



2 mole monoatomic gas

Sol. $W_g = (A \rightarrow B) = 0$ [Isochoric]

$$\Delta U(A \rightarrow B) = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(300)}{2} = 900R$$

(B $\rightarrow$ C) Isothermal

$$W_g(B \rightarrow C) = nRT \ln \frac{V_2}{V_1} = 2 \times R \times 400 \ln \frac{800}{200} = 800R \ln 4$$

$$\Delta U_{(B \rightarrow C)} = 0$$

(C $\rightarrow$ A) Isobaric

$$W_{g(C \rightarrow A)} = nR\Delta T = 2 \times R \times (100 - 400) = -600R$$

$$\Delta U_{(C \rightarrow A)} = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R(-300)}{2} = -900R$$

Net work done

$$(W_g)_{\text{net}} = 800R \ln 4 - 600R$$

Net internal energy change

$$(\Delta U)_{\text{net}} = 0$$

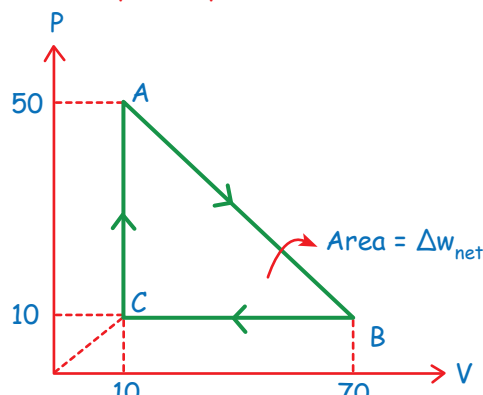
$$\Delta Q_{\text{net}} = \Delta U_{\text{net}} + \Delta W_{\text{net}}$$

$$(\Delta Q)_{\text{net}} = 900R - 1500R + 800R = 200R$$

$$(\Delta Q)_{\text{net}} = \Delta W_{\text{net}} \quad [\Delta U_{\text{net}} = 0]$$

	W_g	ΔU	ΔQ
A $\rightarrow$ B	0	900 R	900 R
B $\rightarrow$ C	$800 R \ln 4$	0	$800 R \ln 4$
C $\rightarrow$ A	-600 R	-900 R	-1500 R

Q. Calculate net heat given to gas in following thermodynamic process ABC.



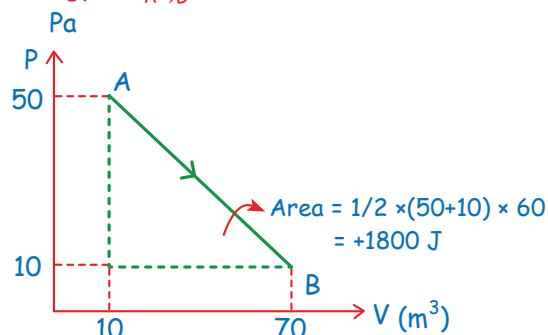
Sol. From 1st law of thermodynamics

$$\Delta Q_{\text{net}} = \Delta U_{\text{net}} + \Delta W_{\text{net}} = \Delta W_{\text{net}}$$

In cyclic process ΔU is 0

$$\frac{1}{2} \times 60 \times 40 = -1200 \rightarrow \text{Area}$$

Q. In the following process A to B, If 500 cal of heat is extracted from gas find change internal energy $\Delta U_{A \rightarrow B}$.



Sol. First law of thermodynamics

$$\Delta Q_{A \rightarrow B} = \Delta U_{A \rightarrow B} + \Delta W_{A \rightarrow B}$$

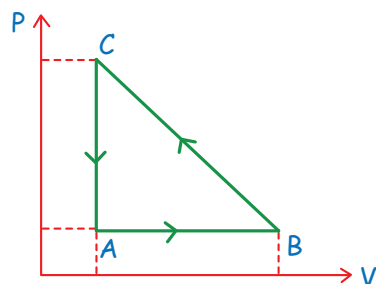
$$-500 = \Delta U_{AB} + 1800$$

$$\Rightarrow \Delta U_{AB} = -2300 \text{ [T} \downarrow \text{] - Temp decrease}$$

$$-2100 = \Delta U_{AB} + 1800 \text{ [500 cal = } 500 \times 4.2 = 2100 \text{ J]}$$

$$\Delta U_{AB} = -3900 \text{ J}$$

Q. In following thermodynamic process 2 mole monoatomic gas at temperature $T_A = 300\text{K}$, $T_B = 500\text{K}$. Find work done by gas $(WD)_g$ in process $B \rightarrow C$. If total 120 J of heat is rejected by gas.



Sol. $W_{AB} = nR\Delta T = 400 \text{ R}$ - isobaric

$$\Delta Q_{\text{net}} = -120 - \text{Given}$$

$$\Delta U_{\text{net}} = 0$$

$$W_{\text{net}} = -120 - [\Delta Q = \Delta W + \Delta U]$$

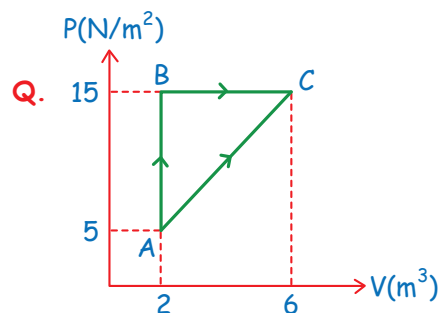
$$W_{\text{net}} = (WD)_{AB} + (WD)_{BC} + (WD)_{CA} = -120$$

$$400\text{R} + (WD)_{BC} + 0 = -120$$

$$(WD)_{BC} = -120 - 400\text{R}$$

$$(WD)_{BC} = -120 - 400\text{R}$$

OR Area under curve [JEE Adv.]



Q. (a) Find (WD) along both the path $A \rightarrow B \rightarrow C$ & $A \rightarrow C$

Sol. $(WD)_g A \rightarrow B \rightarrow C = W(A \rightarrow B) + W(B \rightarrow C)$

$$(WD)_g A \rightarrow C =$$

$$0 + 15 \times 4 = 60$$

$$1/2 (5 + 15) \times 4 = +40$$

(b) $\rightarrow$ If internal energy of gas at A is 20 J and amount of heat supplied to change its state to $^\circ\text{C}$ through path AC is 200 J. Find U_C .

Sol. $U_A = 20\text{J}$

$$\Delta Q_{AC} = +200$$

First law of thermodynamic

$$\Delta Q_{AC} = \Delta U_{AC} + \Delta W_{AC}$$

$$200 = \Delta U_{AC} + 40$$

$$\Rightarrow \Delta U_{AC} = 160$$

$$\Delta U_{AC} = U_C - U_A = 160$$

$$U_C - 20 = 160 \text{ [} U_A = 20 \text{]}$$

$$U_C = 180 \text{ J}$$

(c) $\rightarrow$ If internal energy of gas at state B is 60 J. & at A = 20J. Find the amount of heat supplied to gas from $A \rightarrow B$.

Sol. $U_B = 60$ - Given

First law of thermodynamics

$$\Delta Q_{AB} = \Delta U_{AB} + \Delta W_{AB}$$

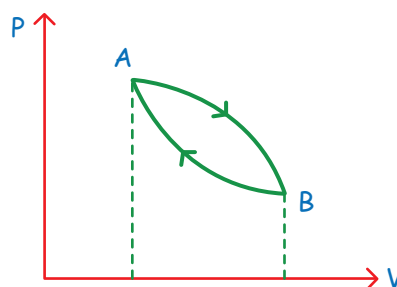
$$\Delta Q_{A \rightarrow B} = U_B - U_A + 0 \text{ [Isochoric process } A \rightarrow B \text{]}$$

$$= 60 - 20 \text{ (} U_A = 20\text{J)}$$

$$= 40\text{J}$$

Q. If heat absorb in process $A \rightarrow B$ is 100 J. and process $B \rightarrow A$ is adiabatic compression for which heat given $\Delta Q = 0$. Work done on the gas from $B \rightarrow A$ is 70J.

(a) Find change in internal energy in process $A \rightarrow B$.



Sol. First law of thermodynamics $\Delta Q = \Delta U + \Delta W$

$$\Delta Q_{AB} + \Delta Q_{BA} = \Delta W_{AB} + \Delta W_{BA} \text{ [} \Delta U_{\text{net}} = 0 \text{]}$$

$$100 + 0 = \Delta W_{AB} - 70$$

$$\Delta W_{AB} = 170$$

(b) Find change in internal energy from A to B.

$$\Delta U_{A \rightarrow B} = ?$$

1st Method (A → B)

Given, $\Delta Q_{AB} = 100 \text{ J}$

$\Delta Q_{BA} = 0$

in $A \rightarrow B$ $\Delta W = -170 \text{ J}$

$\Delta U = A \rightarrow B$

$\Delta Q = \Delta U + \Delta W$

$100 = \Delta U + \Delta W$

$[\Delta W = 170]$

$\Delta U_{AB} = -70$

2nd Method (B → A)

$B \rightarrow A$ (directly)

$\Delta Q_{BA} = \Delta U_{BA} + \Delta W_{BA}$

$0 = \Delta U_{BA} - 70$

$\Delta U_{BA} = 70$

$\Delta U_{AB} = -70$

SKC

Ques. solve karte waqt iss baat ka dhyaan dena hai, ki WD by the gas diya hai, ya WD on the gas hai. Hum zaroorat padne par Single process ke liye bhi 'FLOT' laga sakte hain, aur complete process ke liye bhi."

Where **FLOT** → **First Law of Thermodynamics**

MOLAR SPECIFIC HEAT CAPACITY

Molar specific heat capacity is the amount of heat required to raise the temperature of one mole of a substance by 1 Kelvin (or 1 degree Celsius).

It is expressed as:

$$C = \frac{\Delta Q}{n\Delta T} \quad \text{or} \quad \Delta Q = nC\Delta T$$

Where:

C is the molar specific heat capacity

Q is the heat added

n is the number of moles

ΔT is the change in temperature

Are yaad hai na calorimetry mein padha tha

$$\Delta Q = mS\Delta T$$

specific heat

$$S = \frac{\Delta Q}{m\Delta T}$$

Amount of heat req. to raise the temp. by 1°C of 1 unit mass

Molar specific heat in different processes

$$C = \frac{\Delta Q}{n\Delta T}$$

★ For Isothermal process

$$\Delta T = 0$$

$$C \rightarrow \infty$$

★ For Adiabatic process

$$\Delta Q = 0$$

$$\Rightarrow C \rightarrow 0$$

★ For Isochoric process

$$C = \frac{\Delta Q}{n\Delta T} = \frac{\Delta W + \Delta U}{n\Delta T} = \frac{\Delta U}{n\Delta T} = \frac{nfR\Delta T}{2n\Delta T} = C_v$$

$$C = C_v = \frac{fR}{2}$$

Molar specific heat at const vol^m

$$\therefore C_v = \frac{fR}{2}, \Delta U = \frac{nfR\Delta T}{2} = nC_v\Delta T$$

★ Isobaric process

$$C = \frac{\Delta U + \Delta W}{n\Delta T} = \frac{\Delta U}{n\Delta T} + \frac{\Delta W}{n\Delta T}$$

$$C = \frac{nfR\Delta T}{2n\Delta T} + \frac{nR\Delta T}{n\Delta T} = \frac{fR}{2} + R = C_p$$

Molar specific heat at const. pressure

$$\Rightarrow C_p = C_v + R$$

$$C_p > C_v$$

$$C_p - C_v = R \rightarrow \text{Gas const.}$$

Molar specific heat at const pressure

Molar specific heat at const volume.

C_v mein R jor do C_p ke liye

$$\gamma = \frac{C_p}{C_v} = \frac{C_v + R}{C_v} = 1 + \frac{R}{fR/2}$$

$$\gamma = 1 + 2/f$$

$$C_v = fR/2$$

$$\gamma_{\text{mono}} > \gamma_{\text{dia.}} > \gamma_{\text{tri}}$$

$$\gamma = C_p/C_v$$

MIXTURE OF GAS

Specific heat capacity at constant volume for mixture of gases is given by

$$\rightarrow (C_v)_{\text{mix}} = \frac{n_1 C_{v1} + n_2 C_{v2} + \dots}{n_1 + n_2 + \dots}$$

specific heat capacity at constant pressure for mixture of gases is given by

$$\rightarrow (C_p)_{\text{mix}} = \frac{n_1 C_{p1} + n_2 C_{p2} + \dots}{n_1 + n_2 + \dots}$$

Ratio of specific heat capacity at constant pressure and constant volume of a mixture of gas

$$\rightarrow \gamma_{\text{mix}} = \frac{(C_p)_{\text{mix}}}{(C_v)_{\text{mix}}}$$



Kaam ka Dabba

- ★ $C = \Delta Q / n \Delta T$
- ★ Adiabatic, $C = 0$
- ★ Isothermal, $C = \infty$
- ★ Isochoric, $C = C_v = fR/2$
- ★ Isobaric, $C = C_p = C_v + R$
- ★ $\gamma = C_p / C_v = 1 + 2/f$
- ★ Polytropic, $C = C_v + R/1 - x$
- ★ $\rightarrow (C_v)_{\text{mix}} = \frac{n_1 C_{v1} + n_2 C_{v2} + \dots}{n_1 + n_2 + \dots}$
- ★ $\rightarrow (C_p)_{\text{mix}} = \frac{n_1 C_{p1} + n_2 C_{p2} + \dots}{n_1 + n_2 + \dots}$
- ★ $\rightarrow \gamma_{\text{mix}} = \frac{(C_p)_{\text{mix}}}{(C_v)_{\text{mix}}}$

Q. 2 mole He is mixed with 4 mole oxygen. Find C_p & C_v of mixture & $\gamma = C_p / C_v$

2 mole He + 4 mole O_2

Sol. $C_{v1} = 3R/2$

$C_{v2} = 5R/2$

$C_{p1} = 5R/2$

$C_{p2} = 7R/2$

$$(C_v)_{\text{mix}} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2} = \frac{2 \times 3R/2 + 4 \times 5R/2}{6} = \frac{13R}{6}$$

$$(C_p)_{\text{mix}} = \frac{2 \times 5R/2 + 4 \times 7R/2}{6} = \frac{19R}{2}$$

$$\gamma = \frac{19R/2}{13R/6} = \frac{19}{13}$$

Chalo ab detail mein adiabatic process padhte hain

ADIABATIC PROCESS

★ No heat or energy is exchanged between the system and its surroundings.

★ $\Delta Q = 0$

★ $\Delta W = -\Delta U$

★ $PV^\gamma = \text{Constant}$, (where γ is an adiabatic constant)

★ $P_1 V_1^\gamma = P_2 V_2^\gamma$

★ $PV^\gamma \rightarrow \text{const.} \Rightarrow \frac{nRT}{V} V^\gamma = \text{const.}$

Hence, $TV^{\gamma-1} = \text{const.}$

★ Since, $PV^\gamma \rightarrow \text{const.} \Rightarrow P \left(\frac{nRT}{P} \right)^\gamma \rightarrow \text{const.}$

Hence, $P^{1-\gamma} T^\gamma = \text{const.}$

Work Done by Gas in Adiabatic Process

$P_1 V_1^\gamma = P_2 V_2^\gamma = K$

work done by gas

$$(WD)_g = \int P dv = \int_{v_1}^{v_2} \frac{K}{v^\gamma} dv = K \int_{v_1}^{v_2} v^{-\gamma} dv$$

$$W_g = \frac{P_2 V_2 - P_1 V_1}{1-\gamma} = \frac{nR\Delta T}{1-\gamma} \text{ (Already solve)}$$

M-2

From first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W, \text{ Put } \Delta Q = 0$$

$$\Rightarrow \Delta W = -\Delta U = \frac{-nR\Delta T}{2}$$

$$\Delta W = -n \frac{2}{\gamma-1} \frac{R\Delta T}{2} \quad (f = 2/\gamma-1)$$

$$\Delta W = \frac{nR\Delta T}{\gamma}$$

Q. Initially P_0V_0 given, gas compress to $\frac{V_0}{32}$ adiabatically. Find P_f if gas is diatomic

Sol. $PV^\gamma = \text{const.}$ $\gamma_{\text{dia}} = 7/3$

$$P_i V_i^\gamma = P_f V_f^\gamma$$

$$P_0 V_0^{7/5} = P_f \left(\frac{V_0}{32}\right)^{7/5} \quad \gamma = 1 + \frac{2}{f}$$

$$P_0 V_0^{7/5} = P_f V_0^{7/5} \left(\frac{1}{2^5}\right)^{7/5}$$

$$P_0 = P_f \left(\frac{1}{2}\right)^{7/5}$$

$$P_f = 2^7 P_0 = 128 P_0$$

POLYTROPIC PROCESS

In polytropic process

$$PV^x = \text{const.}$$

$$(WD)_{\text{by gas}} = \frac{nR\Delta T}{1-x}, \quad x \neq 1$$

$$(WD)_{\text{gas}} = \frac{P_f V_f - P_i V_i}{1-x}$$

are agar hum x ki jagah chupke se γ likh dein toh kya hi ho jaayega, maths toh same rahegi

Molar Specific Heat for Polytropic Process

$$C = \frac{\Delta U + \Delta W}{n\Delta T} = \frac{nR\Delta T}{2n\Delta T} + \frac{nR\Delta T}{(1-x)n\Delta T}$$

$$C = \frac{fR}{2} + \frac{R}{(1-x)}$$

$$= C_V + \frac{R}{1-x}$$



Kaam ka Dabba

Table: First Law of Thermodynamics Applied to Different Processes

Process	$\Delta U = nC_V\Delta T$	W	ΔQ
Cyclic	$nC_V\Delta T = 0$	Area of the closed curve	ΔW
Isochoric	$nC_V\Delta T$ (n mole of gas)	0	ΔU
Isothermal	$nC_V\Delta T = 0$	$nRT \log_e \left[\frac{V_f}{V_i} \right]$	ΔW
Adiabatic	$nC_V\Delta T$	$\frac{nR(T_f - T_i)}{1-\gamma}$	0
Isobaric	$nC_V\Delta T$	$P(V_f - V_i)$ $= nR(T_f - T_i)$	$nC_p\Delta T$

Q. In following different processes 2 moles of monoatomic gas undergoes thermodynamic process, if temp. is increase by 100K by keeping pressure constant.

(a) Find heat supply, work done by gas (WD)_{by gas}, and change internal energy ΔU

Sol. Since it's isobaric process

$$\Delta U = \frac{nR\Delta T}{2} = \frac{2 \times 3 \times R \times 100}{2} = 300R$$

Work done by gas

$$(WD)_g = nR\Delta T = 2 \times R \times 100 = 200R$$

$$\text{Heat given, } \Delta Q = 300R + 200R = 500R$$

OR

$$\begin{aligned} \Delta Q &= \Delta U + \Delta W = nC_V\Delta T + nR\Delta T = n(C_V + R)\Delta T \\ &= nC_V\Delta T \Rightarrow 2 \times (C_V + R) \times 100 \quad [C_V = 3R/2] \\ &= 500R \end{aligned}$$

(b) Repeat the work problem if process is ideabatic

Sol. $\Delta Q = 0$

$$\Delta U = \frac{nR\Delta T}{2} = \frac{2 \times 3 \times R \times 100}{2} = 300R$$

$$\Delta Q = \Delta U + \Delta W$$

$$0 = 300R + \Delta W$$

$$\Delta W = -300R$$

Q. For process $P = KV^2$, 2 mole monoatomic gas at 100 K.

If volⁿ become double, find C , W_g , ΔU , ΔQ .

Sol. $PV^2 = \text{const.}$

$$x = -2$$

$$C = \frac{fR}{2} + \frac{R}{1-x} = \frac{3(R)}{2} + \frac{R}{1+2} = \frac{11R}{6}$$

If V_0 becomes $2V_0$

P_0 becomes $4P_0$

T_0 becomes $8T_0$

100K $\rightarrow$ 800 K $[\Delta T = 700 \text{ K}]$

Work done by gas

$$(W)_g = \frac{nR\Delta T}{1-x} = \frac{2 \times R \times 700}{3} = \frac{1400R}{3}$$

Change in internal energy

$$\Delta U = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R \times 700}{2} = 2100R$$

Heat given

$$\Delta Q = \frac{7700R}{3}$$

Q. For process $P = KT^2$, 2 mole $\rightarrow$ monoatomic gas at 100 K.

If volⁿ is double, find C , W_g , ΔU , ΔQ .

Sol. $P = KT^2$

If V becomes double

T becomes $1/2$

P becomes $1/4$

Temp change 100 $\rightarrow$ 50 $[\Delta T = -50]$

$$P = K \left(\frac{PV}{nR} \right)^2 = \frac{KP^2V^2}{n^2R^2} \Rightarrow PV^2 = \frac{n^2R^2}{K}$$

$PV^2 \rightarrow \text{const.}$

$$x = 2$$

Molar specific heat capacity

$$C = C_v + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{1-2} = \frac{R}{2}$$

Change in internal energy

$$\Delta U = \frac{nfR\Delta T}{2} = \frac{2 \times 3 \times R \times (-50)}{2} = -150R$$

Work done by gas

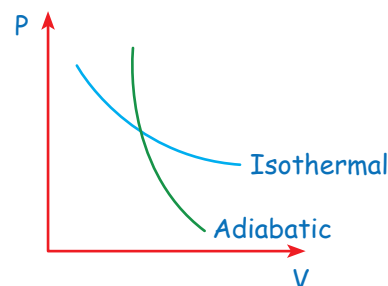
$$\Delta W = \frac{nR\Delta T}{1-x} = \frac{2 \times R \times (-50)}{1-2} = 100R$$

$$\Delta Q = -50R$$

Comparison of P - V Graph in Isothermal Process & Adiabatic Process-

Isothermal	Adiabatic
$PV = \text{const.}$ differentiating w.r.t	$PV^\gamma \rightarrow \text{const}$ differentiating w.r.t
$P \times 1 + V \times \frac{dP}{dV} = 0$	$p^\gamma \gamma V^{\gamma-1} + v^\gamma \frac{dP}{dv} = 0$
$\frac{dP}{dV} = -\frac{P}{V}$	$\gamma \frac{PV^\gamma}{V} + V^\gamma \frac{dP}{dv} = 0$ $\frac{dP}{dV} = -\frac{\gamma P}{V}$ $\gamma = 1 + 2/f, \gamma > 1$

Graph of isothermal & adiabatic



$$\text{Bulk Modulus} = B = \frac{-\Delta P}{\Delta V/V} = \frac{-dP}{dv/V}$$

Bulk modulus of isothermal process

for Isothermal process- $B = P$

for Adiabatic process- $B = \gamma P$

Isothermal Bulk Modules	Adiabatic Bulk modulus
$PV = \text{const.}$	$PV^\gamma \rightarrow \text{const}$
$P \times 1 + \frac{VdP}{dV} = 0$	$P^\gamma \gamma V^{\gamma-1} + v^\gamma \frac{dP}{dv} = 0$
$P = -\frac{VdP}{dV}$	$\gamma \frac{P}{v} + \frac{dp}{dv} = 0$
$B = \frac{-dP}{dV/V} = P$	$-\frac{VdP}{dV} = \gamma P$
	$B = \frac{-dP}{dV/V} = \gamma P$

FREE EXPANSION

In thermodynamics, free expansion refers to a process where a gas expands into a vacuum, resulting in no work done and no heat exchange, leading to a constant internal energy for an ideal gas.

Work done by gas $(WD)_{\text{gas}} = 0$

Since, container and piston are ideal adiabatic

$$\Delta Q = 0$$

$$\Delta Q = \Delta U + \Delta W$$

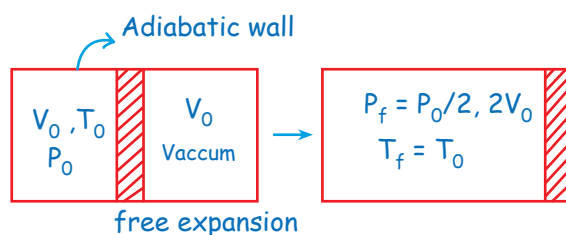
$$0 = \Delta U + 0$$

$$\Rightarrow \Delta U = 0$$

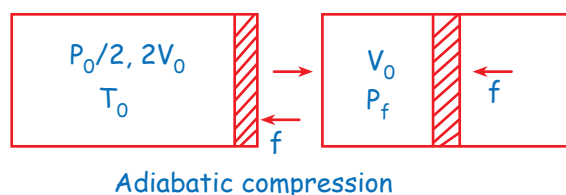
T remain const.

$$T_f = T_0$$

In following free expansion final pressure = $P_0/2$



Q. In the following diagram gas has been compressed adiabatically from initial state $P_0/2, 2V_0, T_0$ to final state P_0, V_0 and T_2



Sol. $P_1 V_1^\gamma = P_2 V_2^\gamma$ $[P_f = P_0 2^{\gamma-1}]$

$$\frac{P_0}{2} (2V_0)^\gamma = P_f V_0^\gamma \quad [T_f = T_0 2^{\gamma-1}]$$

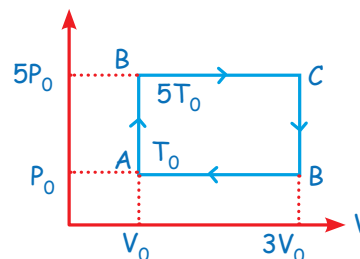
$$T_0 (2V_0)^{\gamma-1} = T_2 (V_0)^{\gamma-1}$$

$$\text{Work done by gas } (WD)_{\text{gas}} = \frac{nR\Delta T}{1-\gamma}$$

→ If piston is brought back to initial position by applying force 'f'. then efficiency η

$$\eta = \frac{(WD)_{\text{net}}}{Q_{+ve}}$$

Q. An ideal gas under goes following process. Find net work done WD_{net} in the process ABCD, & heat transfer ΔQ in each process & efficiency



Sol. Work done = Area of ABCD

$$(WD)_{\text{net}} = 8P_0 V_0$$

heat transfer in following process can be calculated by $\Delta Q = \Delta U + \Delta W$

$$\Delta Q, (A \rightarrow B), \Delta Q > 0$$

$$(B \rightarrow A), \Delta Q > 0$$

$$(C \rightarrow D), \Delta Q < 0$$

$$(D \rightarrow C), \Delta Q < 0$$

$$\text{efficiency } \eta = \frac{8P_0 V_0}{\Delta Q_{AB} + \Delta Q_{BC}}$$

Heat Engine

A heat engine is a device that converts thermal energy (heat) into mechanical work, operating by extracting heat from a hot reservoir, performing work, and then rejecting some heat to a cold reservoir.

Thermal Energy to Work:

Heat engines are designed to transform heat, a form of energy, into mechanical work, which can be used to power machinery or do other tasks.

Cyclic Process:

Heat engines operate in a cycle, meaning they return to their initial state after each cycle, allowing for continuous operation.

Reservoirs:

They involve two reservoirs: a hot reservoir (source of heat) and a cold reservoir (where waste heat is rejected).

Working Substance:

A working substance (like steam, fuel, or air) absorbs heat from the hot reservoir, expands, performs work, and then releases some heat to the cold reservoir.

Efficiency:

The efficiency of a heat engine is the ratio of the work done to the heat absorbed from the hot reservoir.

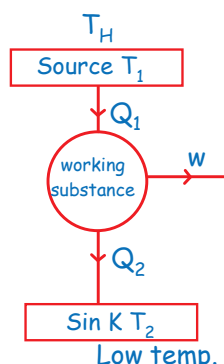
→ Used to convert heat into much work.

$$Q_1 = W + Q_2, \quad W = Q_1 - Q_2$$

$$\eta = \frac{WD}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} = 1 - \frac{T_{\text{small}}}{T_{\text{Big}}}$$

$$= 1 - \frac{T_{\text{sink}}}{T_{\text{source}}} \left[\frac{Q_2}{Q_1} = \frac{T_2}{T_1} \right]$$



Refrigerator [Not in syllabus]

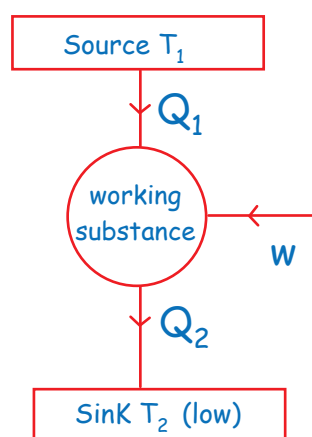
A refrigerator is a device that works on the principle of reversing a heat engine's cycle, extracting heat from a cold reservoir (inside the fridge) and rejecting it to a hotter reservoir (the outside environment), requiring external work.

→ Reverse process of heat engine.

$$W + Q_2 = Q_1$$

Cop = coeff. of performance

$$= \frac{Q_2}{W}$$



2nd Law of Thermodynamics

The second law of thermodynamics, in essence, states that the total entropy of a closed system either increases or remains constant in any spontaneous process; it never decreases. This means heat flows spontaneously from hotter to colder objects, and not the other way around, and not all heat can be converted into work in a cyclic process. 100% conversion of heat is not possible.

ENTROPY

Entropy is a fundamental concept in thermodynamics that quantifies the degree of disorder or randomness in a system. It is a measure of the unavailable energy in a closed system for performing useful work.

The concept of entropy is closely tied to the Second Law of Thermodynamics, which states that the total entropy of an isolated system tends to increase over time. In other words, natural processes in an isolated system lead to an increase in disorder.

Entropy changes occur in various process. If a system gains heat ΔQ from surroundings at a certain temperature T , then the change in entropy,

$$\Delta S = \frac{\text{heat absorbed}}{\text{absolute temperature}}$$

$$\Delta S = \frac{\Delta Q}{T} \Rightarrow \Delta Q = T \Delta S$$

Mathematical form of second law of thermodynamics $\Delta Q = T \Delta S$

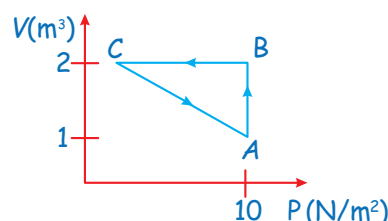
For solids and liquids

The change in entropy during a change of state of mass m of substance $\Delta S = \pm \frac{mL}{T}$

Where (+) sign refers to heat absorption and (-) sign for heat evolution.

Ab main aapke concept aur calculation ko aur better banane ke liye neeche bahut saare question attach kar raha hoon, solve them carefully

Q. An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the work done by the gas in the process $C \rightarrow A$ is



Sol. For cyclic process. Total work done

$$= W_{AB} + W_{BC} + W_{CA}$$

$$\Delta W_{AB} = P \Delta V = 10 (2 - 1) = 10 \text{ J and } \Delta W_{BC} = 0 \text{ (as } V = \text{constant)}$$

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta U = 0 \text{ (Process ABCA is cyclic)}$$

$$\Rightarrow \Delta Q = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$$

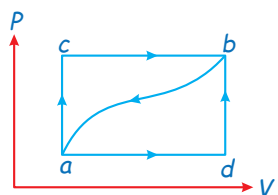
$$\Rightarrow 5 = 10 + 0 + \Delta W_{CA} \Rightarrow \Delta W_{CA} = -5 \text{ J}$$

$$\text{Sol. } \Delta U = Q_{acb} - W_{acb} = 200 - 80 = 120 \text{ J}$$

$$\Delta U = Q_{ba} - W_{ba}, -120 = Q_{ba} + 52,$$

$$Q_{ba} = -172 \text{ J.}$$

Q. When a system is taken from state 'a' to state 'b' along the path 'acb', it is found that a quantity of heat $Q = 200 \text{ J}$ is absorbed by the system and a work $W = 80 \text{ J}$ is done by it. Along the path 'adb', $Q = 144 \text{ J}$. The work done along the path 'adb' is



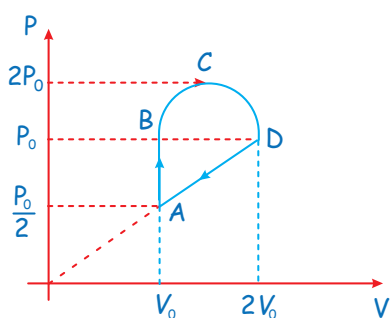
Sol. $\Delta U =$ same in both process

$$Q_{acb} - W_{acb} = Q_{adb} - W_{adb}$$

$$200 - 80 = 144 - W_{adb}$$

$$W_{adb} = 24 \text{ J.}$$

Q. Two moles of a monoatomic ideal gas undergo a cyclic process ABCDA as shown in figure. BCD is a semicircle. Find the efficiency of the cycle.



Sol. Process AB is isochoric ($V = \text{constant}$). Hence

$$\Delta W_{AB} = 0$$

$$\Delta W_{BCD} = P_0 V_0 + \frac{\pi}{2} (P_0) \frac{V_0}{2} = \frac{\pi}{4} + 1 P_0 V_0$$

$$\Delta W_{DA} = -\frac{1}{2} P_0 + P_0 (2V_0 - V_0) = -\frac{3}{4} P_0 V_0$$

$$\Delta U_{AB} = nC_V \Delta T = (2) \frac{3}{2} R (T_B - T_A) \quad \left(n = 2, C_V = \frac{3}{2} R \right)$$

$$= 3R \frac{P_0 V_0}{2R} - \frac{P_0 V_0}{4R} = \frac{3}{4} P_0 V_0 = \Delta Q_{AB} \quad \left(T = \frac{PV}{nR} \right)$$

$$\Delta U_{BCD} = nC_V \Delta T = (2) \frac{3}{2} R (T_D - T_B)$$

$$= (3R) \frac{2P_0 V_0}{2R} - \frac{P_0 V_0}{2R} = \frac{3}{2} P_0 V_0$$

$$\text{Hence, } \Delta Q_{BCD} = \Delta U_{BCD} + \Delta W_{BCD}$$

$$= \frac{\pi}{4} + \frac{5}{2} P_0 V_0 = (2) \frac{3}{2} R (T_A - T_D)$$

$$= (3R) \frac{P_0 V_0}{4R} - \frac{2P_0 V_0}{2R} = -\frac{9}{4} P_0 V_0$$

$$\therefore \Delta Q_{DA} = \Delta U_{DA} + \Delta W_{DA}$$

$$= -\frac{9}{4} P_0 V_0 - \frac{3}{4} P_0 V_0 = -3 P_0 V_0$$

$$\text{Net work done is } W_{\text{net}} = \frac{\pi}{4} + 1 - \frac{3}{4} P_0 V_0 = 1.04 P_0 V_0$$

$$\text{and heat absorbed is } Q_{ab} = \Delta Q_{+ve}$$

$$= \frac{3}{4} + \frac{\pi}{4} + \frac{5}{2} P_0 V_0 = 0.03 P_0 V_0$$

$$\text{Hence efficiency of the cycle is } \eta = \frac{W_{\text{net}}}{Q_{ab}} \times 100$$

$$= \frac{1.04 P_0 V_0}{4.03 P_0 V_0} \times 100 = 25.8\%$$

$$\text{Net work done is } W_{\text{net}} = \frac{\pi}{4} + 1 - \frac{3}{4} P_0 V_0 = 1.04 P_0 V_0$$

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$$= \frac{3}{4} + \frac{\pi}{4} + \frac{5}{2} P_0 V_0 = 0.03 P_0 V_0$$

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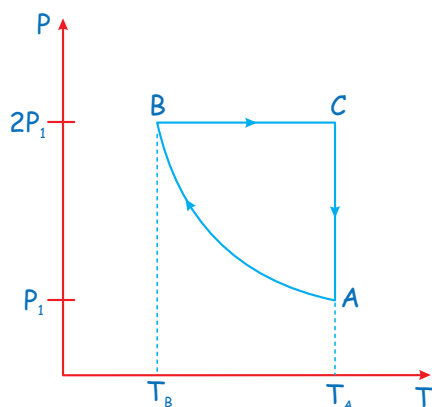
$$= \frac{1.04 P_0 V_0}{4.03 P_0 V_0} \times 100 = 25.8\%$$

Q. Three moles of an ideal diatomic gas is taken through a cyclic process ABCA as shown in the $P - T$ diagram. During the process AB pressure and temperature of the gas varies such that PT^2 is constant. If $T_B = 300 \text{ K}$, calculate:

(a) The work done on the gas in the process AB.

(b) The heat absorbed or released by the gas in each of the processes.

Give answer in terms of the gas constant R .



Sol. (a) $PT^2 = \text{constant}$

$$(2P_1)T_B^2 = (P_1)T_A^2$$

$$T_A = T_B\sqrt{2}$$

$$T_A = T_C = 300\sqrt{2} \text{ K}$$

During the process $A \rightarrow B$

$$PT^2 = C ; P^3V^2 = k$$

$$P = \frac{k^{1/3}}{V^{2/3}}$$

$$W_{AB} = \int_A^B P dV = k^{1/3} \left[\frac{V^{-2/3+1}}{-2/3+1} \right]_{V_A}^{V_B}$$

$$= 3 \left[k^{1/3} \cdot (V_B^{1/3} - V_A^{1/3}) \right]$$

$$W_{AB} = 3(P_B V_B - P_A V_A) = 3nR(T_B - T_A)$$

$$= 9R(300 - 300\sqrt{2}) = 2700R(1 - \sqrt{2})$$

(b) Heat evolved in different processes are

$$Q_{AB} = W_{AB} + \Delta U$$

$$= W_{AB} + nC_V \Delta T$$

$$= \left(9R + \frac{15}{2}R \right) \times 300(1 - \sqrt{2})$$

$$= 150 \times 33R(1 - \sqrt{2}) = -4950R(\sqrt{2} - 1)$$

$$Q_{BC} = nC_p \Delta T = 3 \times \frac{7}{2}R(300R)(\sqrt{2} - 1)$$

$$= 21 \times 150R = 3150R(\sqrt{2} - 1)$$

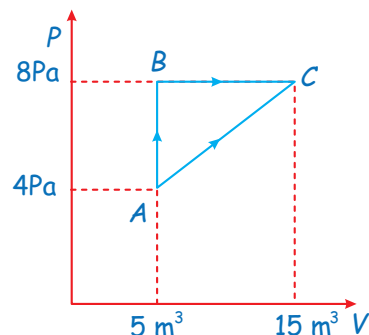
$$Q_{CA} = \Delta U + W = 0 + W_{CA}$$

[As CA is an isothermal process]

$$Q_{CA} = nRT_A \ln \frac{V_A}{V_C} = nRT_A \ln \frac{P_C}{P_A}$$

$$Q_{CA} = 3 \times R \times 300\sqrt{2} \ln(2) = 882.2R$$

Q. In the given figure, an ideal gas changes its state from state A to state C by two paths ABC and AC.



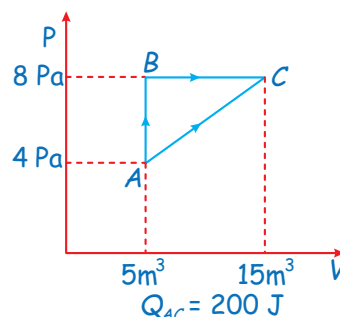
(a) Find the path along which work done is less.

(b) The internal energy of gas at A is 10 J and the amount of heat supplied to change its state to C through path AC is 200 J. Calculate the internal energy of gas at C.

(c) The internal energy of gas at state B is 20 J. Find the amount of heat supplied to the gas to go from A to B.

Sol. (a) W_{AC} is less than W_{ABC} as area under graph is less.

(b) For process A to C



Work done $W_{AC} = \text{area under AC}$

$$= (10 \times 4) + \frac{10 \times 4}{2} = 60 \text{ J}$$

From first law of thermodynamics.

$$\Delta U = Q - W_{AC}$$

$$U_C - U_A = 200 - 60$$

$$U_C = U_A + 140 = 10 + 140 = 150 \text{ J}$$

(c) From A to B

$$U_B = 20\text{J}$$

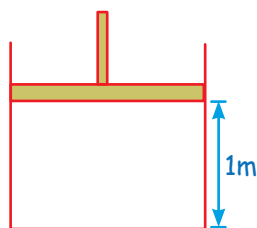
$$\therefore \Delta U = Q - W_{AB}$$

$$U_B - U_A = Q - 0$$

$$20 - 10 = Q$$

$$\therefore Q = 10\text{J}$$

- Q.** The piston cylinder arrangement shown contains a diatomic gas at temperature 300 K. The cross-sectional area of the cylinder is 1 m^2 . Initially the height of the piston above the base of the cylinder is 1 m. The temperature is now raised to 400 K at constant pressure. Find the new height of the piston above the base of the cylinder. If the piston is now brought back to its original height without any heat loss, find the new equilibrium temperature of the gas. You can leave the answer in fraction.



Sol. (a) $4/3\text{ m}$, (b) $T_3 = 400 \left(\frac{4}{3}\right)^{0.4}\text{ K}$

$$\text{Isobaric } \frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{1}{300} = \frac{h}{400}$$

$$\text{Adiabatic } TV^{\lambda-1} = \text{constant.}$$

$$400 (4/3)^{\lambda-1} = T_2 (1)^{\lambda-1}$$

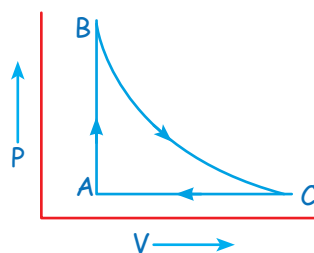
- Q.** One mole of monoatomic ideal gas undergoes a cyclic process ABCA as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450 K respectively. Choose the correct statement(s).

(a) In process CA change in internal energy is 225 R.

(b) In process AB change in internal energy is -150 R.

(c) In process BC change in internal energy -225 R.

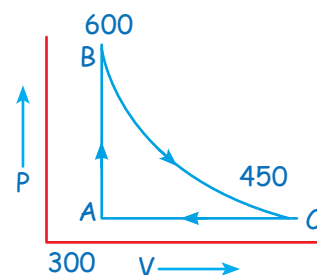
(d) Change in internal energy during the whole cyclic process is +150 R.



Sol. $\Delta U = nC_V \Delta T$ $\Delta T_{AB} = 300$

$$= 1 \times \frac{3}{2} R(\Delta T) \quad \Delta T_{BC} = -150$$

$$\Delta U_{\text{cycle}} = 0 \quad T_{CA} = -150$$



$$\text{or } \Delta T = \frac{2}{3} \times 300 \times 0.693 = 138.6\text{K}$$

This is the change in the temperature of the gas of second vessel.

Now the final temperature of the gas in second vessel is

$$T' = T + \Delta T = 300 + 138.6 = 438.6\text{K}$$

Let after mixing, T_f and P_f be the final temperature and pressure respectively. Then as moles are equal, we have

$$T_f = \frac{T + (T + \Delta T)}{2} = \frac{300 + 438.6}{2}$$

$$= 369.3\text{ K}$$

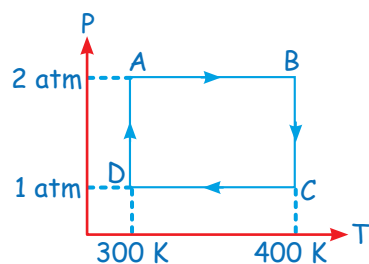
For a gas system from gas law, we have

$$P_f V_f = nRT_f$$

$$\text{or } P_f = \frac{nRT_f}{V_f} = \frac{2 \times 8.3 \times 369.3}{2 \times 8.3 \times 10^{-3} + 8.3 \times 10^{-3}}$$

$$= 2.46 \times 10^5 \text{ N/m}^2$$

- Q.** Two moles of helium gas undergo a cyclic process as shown in figure-3.38. Assuming the gas to be ideal, calculate the following quantities in this process.



- The net change in the heat energy.
- The net work done.
- The net change in internal energy.

Sol. As we know in a cyclic process, the change in heat energy or heat supplied to the gas is equal to the net work done by the gas.

Here AB is isobaric process. Hence work done during this process from A to B is

$$W_{AB} = P(V_2 - V_1) = nR(T_2 - T_1)$$

$$\text{or } W_{AB} = 2 \times 8.314 \times (400 - 300) = 1662.8 \text{ joule}$$

Work done during isothermal process from B to C

$$W_{BC} = nRT_C \ln(V_2/V_1) = nRT_C \ln(P_1/P_2)$$

$$= 2 \times 8.314 \times 400 \times \ln(2)$$

$$= 2 \times 8.314 \times 400 \times 0.693 = 4610.2 \text{ joule}$$

Workdone during isobaric process from C to D

$$W_{CD} = nR(T_D - T_C) = 2 \times 8.314 \times (300 - 400)$$

$$= -1662.8 \text{ joule}$$

Workdone during isothermal process from D to A

$$W_{DA} = nRT_D \ln(P_D/P_A) = nRT_D \ln(2)$$

$$= -2 \times 8.314 \times 300 \times 0.693 = -3457.7 \text{ joule}$$

Net workdone

$$= W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= 1662.8 + 4610.2 - 1662.5 - 3457.7 = 1152.5 \text{ joule}$$

Now from first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

Here $\Delta U = 0$, thus we have

$$\Delta Q = \Delta W = 1152.5 \text{ joule}$$

So the heat given to the system is 1152.5 joule

As the gas returns to its original state, hence there is no change in internal energy.

Q. Two moles of helium gas ($\lambda = 5/3$) are initially at temperature 27°C and occupy a volume of 20 litres. The gas is first expanded at constant pressure until the volume is doubled. Then it undergoes an adiabatic change until the temperature returns to its initial value.

(i) Sketch the process on a P - V diagram.

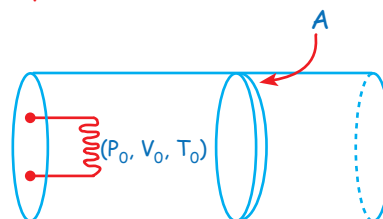
(ii) What are the final volume and pressure of the gas?

(iii) What is the work done by the gas?

Sol. For a perfect gas

$$PV = nRT$$

Q. An ideal gas is contained in a cylinder with a movable piston of cross-sectional area (A). Initially, the gas is at pressure (P_0), volume (V_0), and temperature (T_0). The gas is heated slowly, increasing its temperature to ($2T_0$), while the pressure remains constant at (P_0). Calculate the displacement of the piston during this expansion.



Sol. The process is isobaric (P_0 is constant). Using the ideal gas law, $\frac{V}{T} = \text{constant}$ at constant pressure.

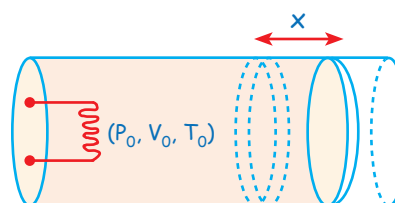
$$\text{Initial state: } \frac{V_0}{T_0}$$

$$\text{Final state: } \frac{V_f}{T_f} \text{ where } T_f = 2T_0$$

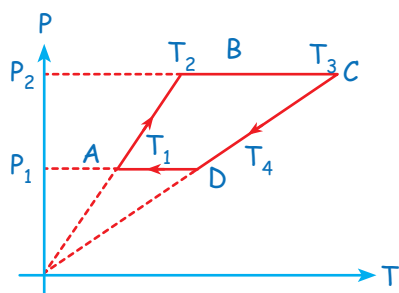
$$\text{Solving: } \frac{V_0}{T_0} = \frac{V_f}{2T_0}$$

$$\text{The change in volume is } \Delta V = V_f - V_0 = V_0$$

$$\text{Since } \Delta V = Ax, \text{ we have } x = \frac{\Delta V}{A} = \frac{V_0}{A}$$



Q. An ideal gas undergoes a cyclic process as shown in the P-T diagram below. If the gas has (n) moles, calculate the total work done by the gas in the cycle.



Sol. Isochoric processes (AB, CD): ($W = 0$).

For isobaric processes:

• BC: $W_{BC} = P\Delta V = nR\Delta T = nR(T_3 - T_2)$.

• DA: $W_{DA} = nR(T_1 - T_4)$.

Total work done:

$$W = W_{BC} + W_{DA} = nR(T_3 - T_2 + T_1 - T_4).$$

Q. One mole of an ideal gas expands such that its volume is proportional to $T^{2/3}$, where T is the temperature. If the temperature of the gas increases by 30°C during this process, calculate the work done by the gas.

Sol. For one mole of an ideal gas, $PV = RT$, so $P = \frac{RT}{V}$.

Given $V \propto T^{2/3}$, let $V = kT^{2/3}$.

Then, $P = \frac{RT}{kT^{2/3}} = \frac{RT^{1/3}}{k}$.

Differential work: $dW = PdV$.

Since $V = kT^{2/3}$ $dV = k \cdot \frac{2}{3} T^{-1/3} dT$.

Thus, $PdV = \left(\frac{R}{k} T^{1/3}\right) \left(k \cdot \frac{2}{3} T^{-1/3} dT\right) = \frac{2R}{3} dT$.

$$\Rightarrow W = \int_{T_i}^{T_f} \frac{2R}{3} dT = \frac{2R}{3} (T_f - T_i).$$

Given $T_f - T_i = 30\text{K}$, $W = \frac{2R}{3} \times 30 = 20R$

Q. One mole of an ideal gas with adiabatic exponent γ undergoes a process where its volume varies as $V = \frac{a}{T}$, with a constant. If the temperature of the gas increases by ΔT , calculate the heat absorbed by the gas during this process.

Sol. For one mole, ($PV = RT$). Given $V = \frac{a}{T} \Rightarrow P = \frac{RT^2}{a}$.

$$dV = -\frac{a}{T^2} dT.$$

Work done: $dW = PdV = \left(\frac{RT^2}{a}\right) \left(-\frac{a}{T^2} dT\right) = -RdT$.

$$W = \int_T^{T+\Delta T} -RdT = -R\Delta T$$

Internal energy change: $\Delta U = \frac{R}{\gamma-1} \Delta T$.

By the first law, $Q = \Delta U + W$:

$$Q = \frac{R}{\gamma-1} \Delta T - R\Delta T = R\Delta T \left(\frac{1}{\gamma-1} - 1\right) = R\Delta T \left(\frac{2-\gamma}{\gamma-1}\right)$$

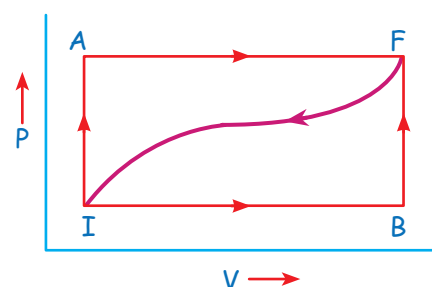
Q. A system transitions from state I to state F via path IAF, absorbing ($Q = 55\text{ J}$) of heat and doing ($W = 25\text{ J}$) of work, as shown in the P-V diagram. Along path IBF, ($Q = 35\text{ J}$).

(a) Calculate the work done along the IBF path.

(b) If $W = -15\text{ J}$ along path FI, how much heat is lost?

(c) If $U_I = 10\text{ J}$, find U_F .

(d) If $U_B = 20\text{ J}$, find Q for processes BF and IB.



Sol. Using $Q = \Delta U + W$, for IAF: $\Delta U = 55 - 25 = 30\text{ J}$.

(a) Along IBF: $W = 35 - 30 = 5\text{ J}$.

(b) Along FI: $\Delta U = -30\text{ J}$, $Q = -30 - 15 = -45\text{ J}$.
Heat lost = 45 J .

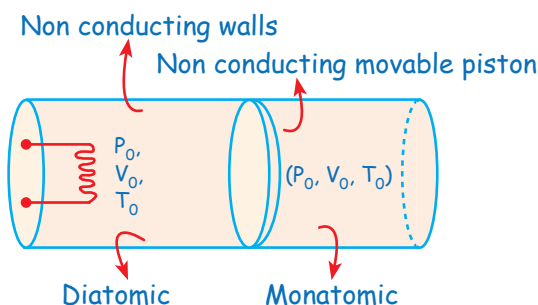
(c) $U_F = 10 + 30 = 40\text{ J}$.

(d) BF (isochoric): $Q_{BF} = 40 - 20 = 20\text{ J}$.

IB (Isobaric): $Q_{IB} = 35 - 20 = 15\text{ J}$.

Q. A cylindrical container with nonconducting walls is divided into two equal parts, each of volume V_0 , by a nonconducting movable piston. The left part holds a diatomic gas, and the right part holds a monatomic gas, both initially at P_0, V_0 ,

T_0 , with n moles in each part. The left gas is heated, compressing the right gas to a volume $\frac{V_0}{8}$. Find the final pressure and temperature on both sides, and the heat supplied to the left gas.



Sol. The process is adiabatic for the right gas ($PV^\gamma = \text{constant}$, $\gamma = \frac{5}{3}$):

$$P_0 V_0^\gamma = P_{\text{final}} \left(\frac{V_0}{8} \right)^\gamma$$

$$P_{\text{final}} = P_0 \left(\frac{V_0}{\frac{V_0}{8}} \right)^{\frac{5}{3}} = 32P_0$$

$$\text{For temperature: } \frac{T_0 V_0^{\gamma-1}}{T_{\text{final}}} = \left(\frac{V_0}{8} \right)^{\gamma-1}$$

$$T_{\text{final}} = T_0 (8)^{2/3} = 4T_0$$

$$\text{Volume of left part: } 2V_0 = V_1 + \frac{V_0}{8}, \text{ so } V_1 = \frac{15V_0}{8}$$

$$\text{For the left gas } \left(\frac{PV}{T} = \text{constant} \right):$$

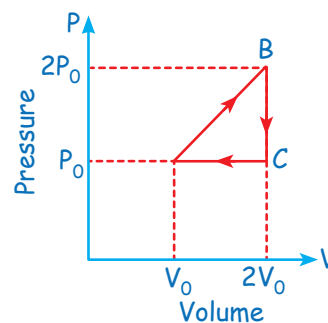
$$\frac{P_0 V_0}{T_0} = \frac{P_{\text{final}} V_1}{T_{\text{final}}}$$

$$T_{\text{final}} = \frac{32P_0 \left(\frac{15V_0}{8} \right)}{P_0 V_0} T_0 = 60T_0$$

$$\text{Heat supplied to left gas } \left(\gamma = \frac{7}{5} \right):$$

$$\Delta Q = n \frac{5R}{2} (60T_0 - T_0) + n \frac{3R}{2} (4T_0 - T_0) = 152nRT_0$$

Q. An ideal diatomic gas undergoes a thermodynamic cycle as shown in the figure. Calculate the efficiency of the cycle.



Sol. Let n be the number of moles of the gas and temperature at point A is T_0 .

Work done in the cycle (area of triangle):

$$W = \frac{1}{2} (2V_0 - V_0) (2P_0 - P_0) = \frac{1}{2} P_0 V_0$$

$$\text{AB (linear): } W_{AB} = \text{area under AB} = \frac{1}{2} (P_0 + 2P_0) (2V_0 - V_0) = \frac{3}{2} P_0 V_0$$

$$\text{At A: } P_0 V_0 = nRT_0. \text{ At B: } T_B = 4T_0$$

$$\Delta U_{AB} = nC_V \Delta T = n \frac{5R}{2} (4T_0 - T_0) = \frac{15}{2} nRT_0$$

$$Q_{AB} = W_{AB} + \Delta U_{AB} = \frac{3}{2} P_0 V_0 + \frac{15}{2} nRT_0 = \frac{9}{2} P_0 V_0$$

$$\text{BC (isochoric, } W = 0): Q_{BC} = n \frac{5R}{2} (2T_0 - 4T_0) = -5nRT_0$$

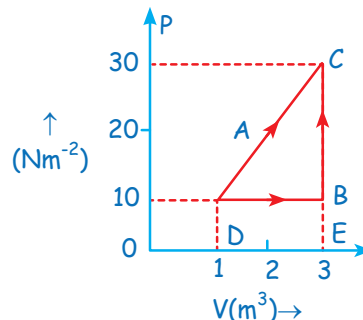
$$\text{CA (isobaric): } Q_{CA} = n \frac{7R}{2} (T_0 - 2T_0) = -\frac{7}{2} nRT_0$$

$$\text{Total heat supplied (positive Q): } Q_{AB} = \frac{9}{2} P_0 V_0$$

Efficiency:

$$\eta = \frac{W}{Q_{\text{supplied}}} \times 100 = \frac{\frac{1}{2} P_0 V_0}{\frac{9}{2} P_0 V_0} \times 100 = \frac{50}{9} \%$$

Q. An ideal gas is taken round a cyclic thermodynamic process ABCA as shown in the given figure. If the internal energy of gas at point A is assumed zero while at B it is 50 J. The heat absorbed by the gas in the process BC is 90 J.



- (a) What is the internal energy of the gas at point C?
- (b) How much heat energy is absorbed by the gas in the process AB?
- (c) Find the heat energy rejected or absorbed by the gas in the process CA.
- (d) What is the net work done by the gas in the complete cycle ABCA?

Sol. (a) BC (isochoric, $W_{BC} = 0$):

$$Q_{BC} = \Delta U_{BC} = U_C - U_B.$$

$$90 = U_C - 50, \text{ so } U_C = 140 \text{ J.}$$

(b) AB is Isobaric process:

$$W_{AB} = \text{Area under the curve in this process}$$

$$= 10 \times (3 - 1) = 20 \text{ J}$$

$$Q_{AB} = \Delta U_{AB} + W_{AB} = 50 + 20 = 70 \text{ J}$$

(c) CA Process:

$$W_{CA} = \text{Area of trapezium ACED}$$

$$= \frac{1}{2} \times (30 + 10) \times (1 - 3) = -40 \text{ J}$$

$$Q_{CA} = \Delta U_{CA} + W_{CA} = U_A - U_C + W_{CA}$$

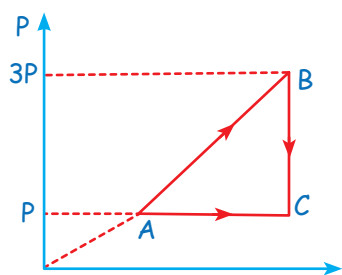
$$= 0 - 140 - 40 = -180 \text{ J}$$

(d) Net work done = area of triangle ABC

$$= \frac{1}{2} \times 2 \times 20 = 20 \text{ J}$$

Since cycle is anticlockwise therefore net work done by the gas is 20 J.

Q. A fixed mass of oxygen gas performs a cyclic process ABCA as shown. Find the efficiency of the process.



Sol. AB (isochoric, $W_{AB} = 0$):

$$Q_{AB} = nC_V(3T_A - T_A) = 5nRT_A.$$

$$\text{BC (isothermal): } W_{BC} = nR(3T_A) \ln\left(\frac{P_B}{P_C}\right) = 3nRT_A \ln 3.$$

$$= Q_{BC}$$

$$\text{CA: } W_{CA} = nR(T_A - 3T_A) = -2nRT_A.$$

$$\text{Net work: } W = W_{BC} + W_{CA} = 3nRT_A \ln 3 - 2nRT_A.$$

Efficiency of a cyclic process is given by

$$\eta = \frac{\text{Work done in cyclic process}}{\text{Heat supplied to the system}} = \frac{W}{Q_{in}}$$

$$\text{Efficiency, } \eta = \frac{3 \ln 3 - 2}{5 + 3 \ln 3} \% = \frac{3 \ln 3 - 2}{5 + 3 \ln 3} \%$$

Q. In moles of a diatomic gas has undergone a cyclic process ABC as shown in figure. Temperature at A is T_0 .

(a) Find volume at C.

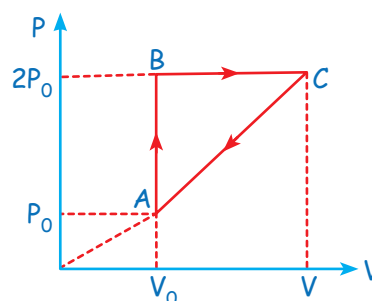
(b) Find maximum temperature.

(c) Find total heat given to gas.

(d) Is heat rejected by the gas.

If yes, how much heat is rejected?

(e) Find out the efficiency.



Sol. (a) Volume at C

$$\frac{2P_0}{V} = \frac{P_0}{V_0} \Rightarrow V = 2V_0$$

$$\text{(b) AB (isochoric): } \frac{P_A}{T_A} = \frac{P_B}{T_B} \cdot \frac{P_0}{T_0} = \frac{2P_0}{T_B},$$

$$\text{so } T_B = 2T_0.$$

$$\text{BC (isobaric): } \frac{V_B}{T_B} = \frac{V_C}{T_C} \cdot \frac{V_0}{2T_0} = \frac{2V_0}{T_C}, \text{ so } T_C = 4T_0.$$

$$\text{Maximum temperature: } T_C = 4T_0.$$

(c) Heat supplied:

AB (isochoric, $W_{AB} = 0$):

$$Q_{AB} = nC_V(T_B - T_A) = n \frac{5R}{2} (2T_0 - T_0) = \frac{5}{2} nRT_0$$

$$\text{BC (isobaric): } Q_{BC} = nC_P(T_C - T_B)$$

$$= n \frac{7R}{2} (4T_0 - 2T_0) = 7nRT_0.$$

(d) Total heat supplied (Q_{in}):

$$Q_{AB} + Q_{BC} = \frac{5}{2} nRT_0 + 7nRT_0 = \frac{19}{2} nRT_0.$$

$\therefore \Delta Q$ is negative in process CA and heat is rejected in process CA

$$\begin{aligned}
 \Delta Q_{CA} &= \Delta W_{CA} + \Delta U_{CA} \\
 &= -\frac{1}{2}[P_0 + 2P_0]V_0 - \frac{5}{2}nR(T_C - T_a) \\
 &= -\frac{1}{2}[P_0 + 2P_0]V_0 - \frac{5}{2}nR\left(\frac{4P_0V_0}{nR} - \frac{P_0V_0}{nR}\right) \\
 &= -9P_0V_0 = \text{Heat injected.}
 \end{aligned}$$

(e) Net work:

$$W = \Delta W_{CA} = -\frac{1}{2}P_0V_0. \text{ Work by gas: } \frac{1}{2}P_0V_0.$$

Efficiency:

$$\eta = \frac{W}{Q_{in}} \times 100 = \frac{\frac{1}{2}P_0V_0}{\frac{19}{2}RT_0} \times 100 = \frac{1}{19} \times 100 \approx 5.26\%$$

Q. A Carnot refrigerator, using a perfect gas, operates between a freezer at 0°C and an outside temperature of 20°C , driven by a 200 W motor. Calculate the mass of water frozen per minute, given the specific latent heat of fusion of ice as 80 cal/g and $J = 4.2 \text{ J/cal}$.

Sol. Coefficient of performance (β) for a Carnot refrigerator:

$$\beta = \frac{T_2}{T_1 - T_2} = \frac{273}{293 - 273} = \frac{273}{20}.$$

Work done per second (W): 200 J/s.

Heat extracted per second (Q_2): $\beta = \frac{Q_2}{W}$,

$$\text{so } Q_2 = \beta W = \frac{273}{20} \times 200 = 2730 \text{ J/s.}$$

Heat extracted per minute: 2730×60
 $= 163,800 \text{ J/min.}$

Convert to calories: $\frac{163,800}{4.2} = 39,000 \text{ cal/min}$

Mass of water frozen per minute (m):

$$m \times 80 = 39,000,$$

$$\text{so } m = \frac{39,000}{80} = 487.5 \text{ g/min.}$$